Activity 8

HackerRank Algorithmic Problem-Solving & Portfolio Integration

Name: Ram Chougala

Course: B.Tech CSE - 3rd Semester

Activity: HackerRank Algorithmic Problem-Solving & Portfolio Integration

GitHub Repository

https://github.com/ramchougala2/HackerRank-3rdSem-Portfolio

HackerRank Profile

Ram Chougala

Repository Overview

The GitHub repository HackerRank-3rdSem-Portfolio contains C++ solutions for five HackerRank algorithmic problems. The repository objective is to practice algorithmic problem solving, understand efficient techniques, analyze time and space complexity, and maintain the solutions as a portfolio.

Problems Solved

Problem	Topic	Time	Space
Diagonal Difference	2D Arrays / Matrices	O(N)	O(1)
Dynamic Array	Data Structures / Vectors	O(N + Q)	O(N)
Time Conversion	Strings / Logic	O(1)	O(1)
Compare the Triplets	Arrays / Comparison	O(1)	O(1)
Sparse Arrays	Strings / Hash Maps	O(N + Q)	O(N)

Problem Details

• Diagonal Difference: Computes the absolute difference between the sums of the two diagonals of a square matrix.

• Dynamic Array: Uses dynamic sequences and XOR-based indexing to process queries.

• Time Conversion: Converts a 12-hour AM/PM time format into 24-hour format.

• Compare the Triplets: Compares Alice's and Bob's scores position by position and calculates their respective points.

• Sparse Arrays: Counts how many times each query string occurs in the given string list.

Complexity Analysis

The following complexity values are taken from the repository README and summarize the efficiency of the five solutions.

Problem	Time Complexity	Space Complexity
Diagonal Difference	O(N)	O(1)
Dynamic Array	O(N + Q)	O(N)
Time Conversion	O(1)	O(1)
Compare the Triplets	O(1)	O(1)
Sparse Arrays	O(N + Q)	O(N)

Repository Structure

```
HackerRank-3rdSem-Portfolio/
├── README.md
├── Diagonal-Difference/solution.cpp
├── Dynamic-Array/solution.cpp
├── Time-Conversion/solution.cpp
├── Compare-the-Triplets/solution.cpp
└── Sparse-Arrays/solution.cpp
```

Reflection

This activity provided practical experience in solving algorithmic problems using C++. The five problems covered arrays, matrices, vectors, strings, hashing, and query-based processing.

The work also emphasized selecting an appropriate data structure and considering time and space complexity while designing a solution. Problems such as Diagonal Difference and Compare the Triplets strengthened direct array-based reasoning, while Dynamic Array and Sparse Arrays provided practice with data structures and efficient query handling.

Maintaining the solutions in a GitHub repository helped organize the code as a portfolio and made the learning work easier to review. Overall, the activity improved algorithmic thinking, implementation practice, and awareness of writing solutions that are both correct and efficient.

Source: GitHub repository maintained by Ram Chougala - HackerRank-3rdSem-Portfolio.